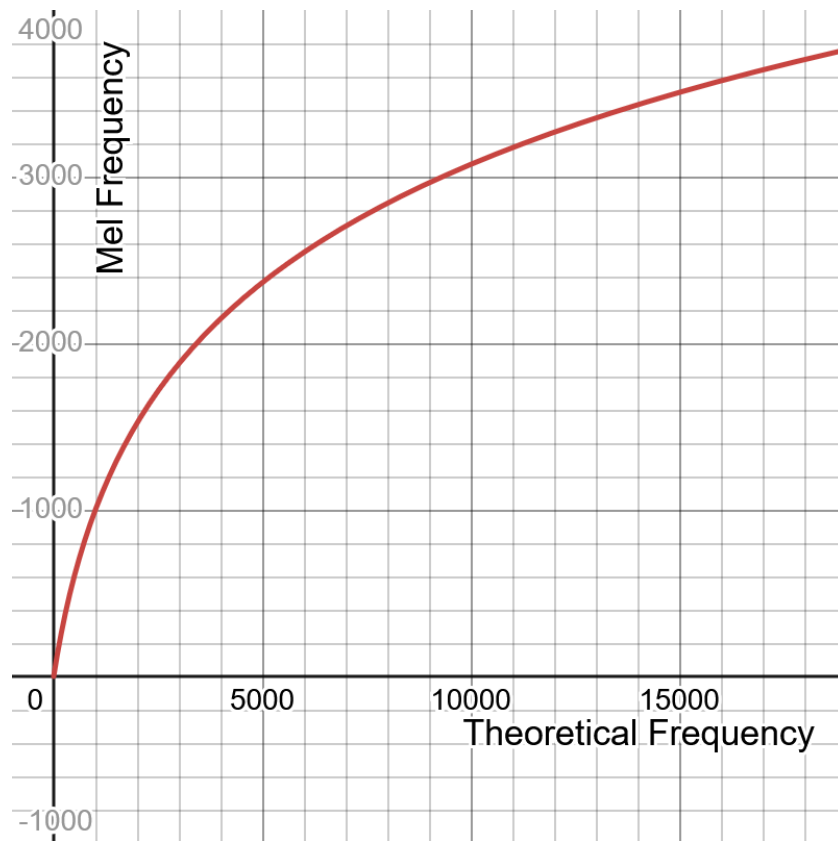


Mel Spectrogram

Motivation

We humans have a logarithmic perception scale when it comes to audio. We can discern two sounds of say $\Delta f = 200\text{Hz}$ very clearly when we are working with sounds on the lower end of the frequency scale but we cannot do the same when working at the higher end of the scale. This happens because the way we perceive frequency follows a logarithmic scale as shown below –



The perception scale equivalent of audio frequency is called Mel frequency scale where the word Mel comes from *melody*. There is an empirical formula to map theoretical frequency, f , with Mel frequency, m as given by –

$$m = 2595 \cdot \log_{10}\left(1 + \frac{f}{700}\right)$$

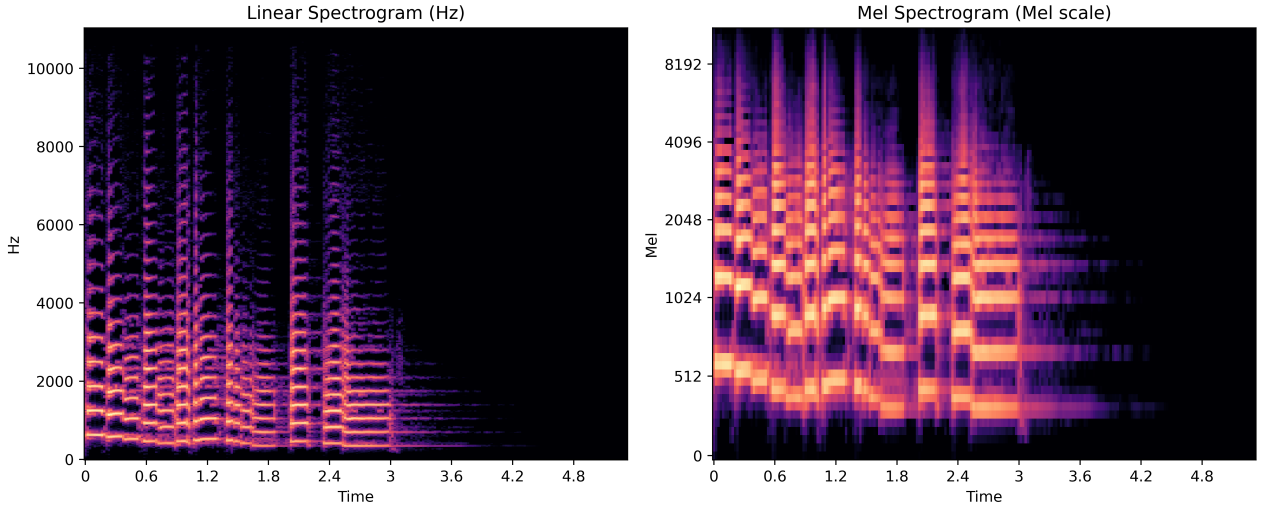
An inverse relationship to convert m back to f can be established by rearranging the formula –

$$f = 700 \cdot (10^{m/2595} - 1)$$

Now, the question is - *what is a Mel spectrogram and why is it important?*

Mel spectrogram is a Mel conversion of a general spectrogram. It is important because through Mel spectrogram you can literally encode human hearing as an image and as such it is a staple in AI-based audio processing.

Mel spectrogram and the theory behind conversion



While it may be tempting to do a one to one mapping of every frequency bin in a regular spectrogram to a Mel spectrogram, its not so straightforward. Lets build to the problem slowly. Firstly, it should be noted that we want our frequency axis to be uniformly sampled, be it in Hz or Mel . In a regular spectrogram, this is ensured in the Hz scale through our Fourier formulation. But since the Mel scale is a transformed scale that *we* generate, we have to ensure uniformity by taking equally spaced discrete points in between a max, m_{max} , and a min, m_{min} , value.

To keep the parallelism intact between the two scales, we take $m_{max} = \text{Mel conversion}(f_{max})$ and $m_{min} = \text{Mel conversion}(f_{min})$. Assume that there is a spectrogram where $f_{min} = 0kHz$ and $f_{max} = 16kHz$. Then,

$$m_{min} = 2595 \cdot \log_{10}\left(1 + \frac{f_{min}}{700}\right) = 2595 \cdot \log_{10}\left(1 + \frac{0}{700}\right) = 0 \text{ Mel}$$

$$m_{max} = 2595 \cdot \log_{10}\left(1 + \frac{f_{max}}{700}\right) = 2595 \cdot \log_{10}\left(1 + \frac{16000}{700}\right) = 3574.92 \text{ Mel} \approx 3575 \text{ Mel}$$

Note: We will round decimals to integers while working with Mel spectrograms for the sake of simplicity.

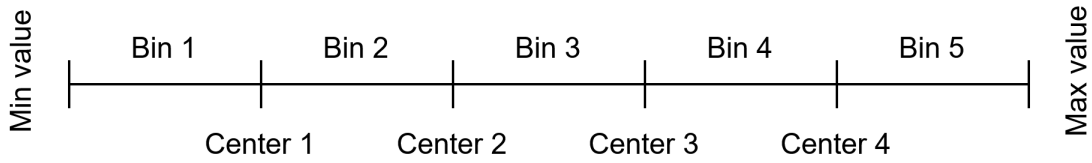
Now, having calculated our m_{min} and m_{max} , we have to divide the range into equally spaced M discrete centers. Given the difference between each centers is,

$$\Delta m = \frac{m_{max} - m_{min}}{M + 1}$$

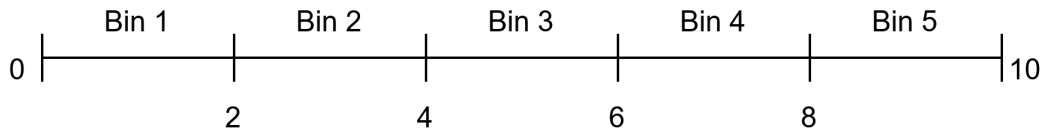
The centers will be $\{m_{min} + \Delta m, m_{min} + 2 \cdot \Delta m, \dots, m_{min} + M \cdot \Delta m\}$.

Do note that the number of centers, M , taken is a design choice. If M is high, your Mel spectrogram will have higher frequency (Mel) resolution. But keep in mind that number of Mel points, M , should be kept less than the number of linear frequency points. At best you can make them equal. You will realize the reason behind this on your own when you complete the theory behind the conversion.

For visual verification of the formula for Δm and the center value formulation, check out the figure given below –



Example -



$$\text{Bin length} = \frac{\text{max}-\text{min}}{n(\text{centers})+1}$$

Note: The bins mentioned in the figure should not be confused with bins written elsewhere in the text. The word 'bin' has been used in the figure for lack of better words describing the same thing.

Continuing with the example from before, lets consider $M = 4$. Then,

$$\Delta m = \frac{m_{\text{max}} - m_{\text{min}}}{M + 1} = \frac{3575}{5} = 715$$

Then the centers will be {715, 1430, 2145, 2860}.

Now if we try to map them one to one with the linear frequency, we will need the equivalent linear frequencies. Using the Mel to linear conversion equation, we get the adjacent linear frequency centers as - {620, 1790, 3996, 8156} [All values have been rounded to integers].

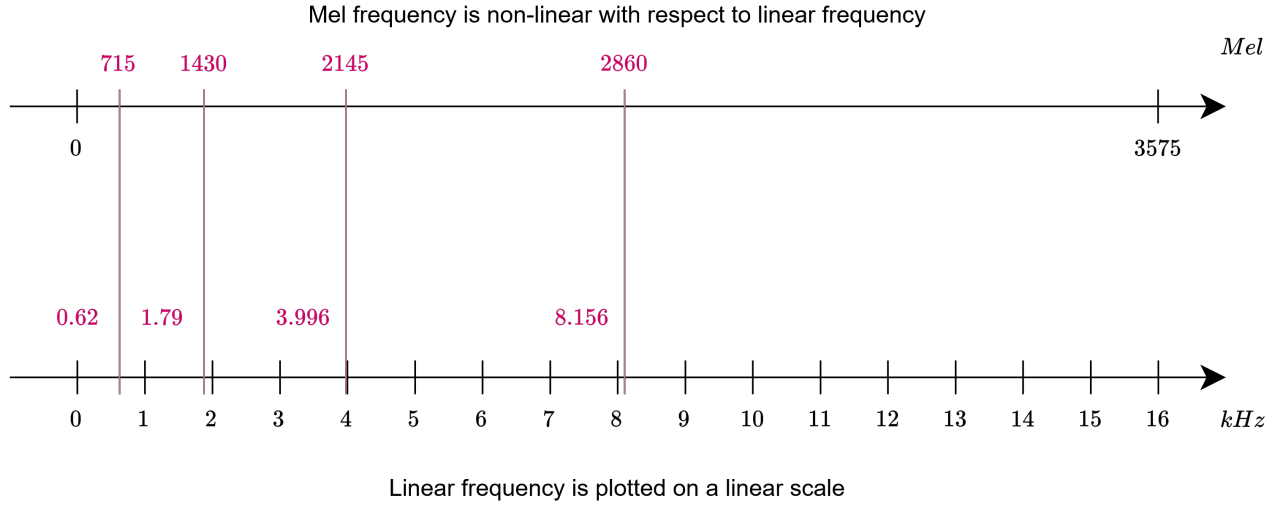
Now it would be very cool, if we have these values in our original spectrograms, but in most cases we will not. Our spectrograms give us discrete frequency information and so not every frequency information is available to us. Additionally, in accurate implementations, we do not round our frequency values, which makes it even more improbable to get a one to one map.

If one to one mapping is not possible, what can we do?

We will use neighborhood information to estimate the frequency component. This is done by creating Mel filter-banks.

Mel Filter-Banks

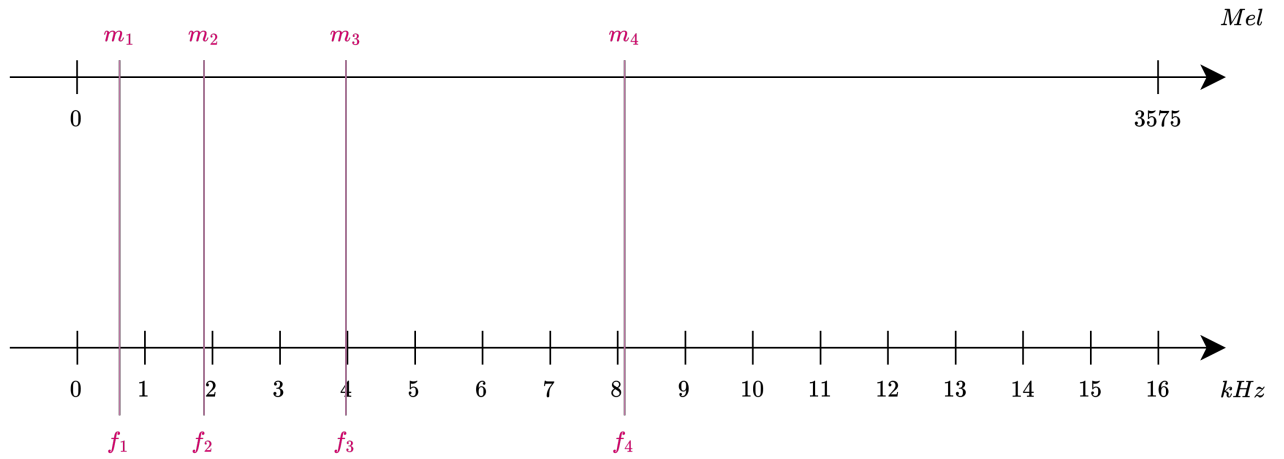
Continuing with the example, let's assume our original spectrogram had a frequency resolution of $\Delta f = 1 \text{ kHz}$. With this information let's map linear frequencies to Mel frequencies on two parallel number lines –



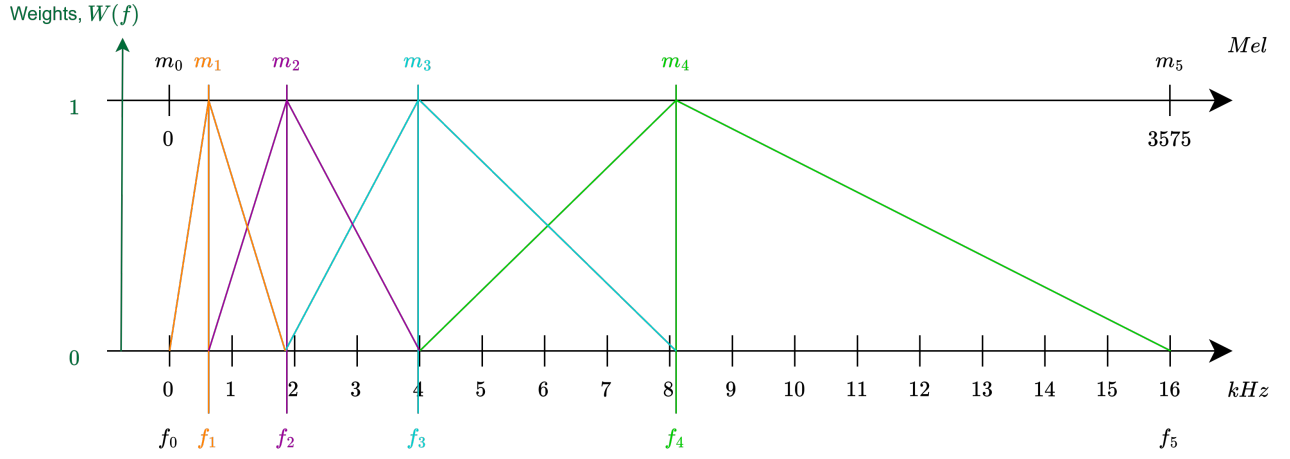
Notice that the linear frequency is plotted on a linearly spaced number line. The adjacent Mel frequency number line is non-linear for one to one mapping as the mapping is logarithmic in nature.

Also note how the one to one mapping misses available discrete frequencies on the linear scale.

For generalized cases, we will replace the numbers with indexed variables in the same figure before proceeding to prepare the frequency neighborhoods –



Now each, m_i will get its own weighted neighborhood also termed as a Mel filter (we will denote each filter as W_i , shorthand for weight). In literature these neighborhoods are weighted using a triangle filter that starts from f_{i-1} , peaks at f_i and continues till f_{i+1} . For the ease of notation we will refer to the lower cut-off frequency as F_{lower}^i , center frequency as F_{center}^i and upper cut-off frequency as F_{upper}^i . The i th superscript indicates the index of the filter. Let's draw the filters in the same plot again –

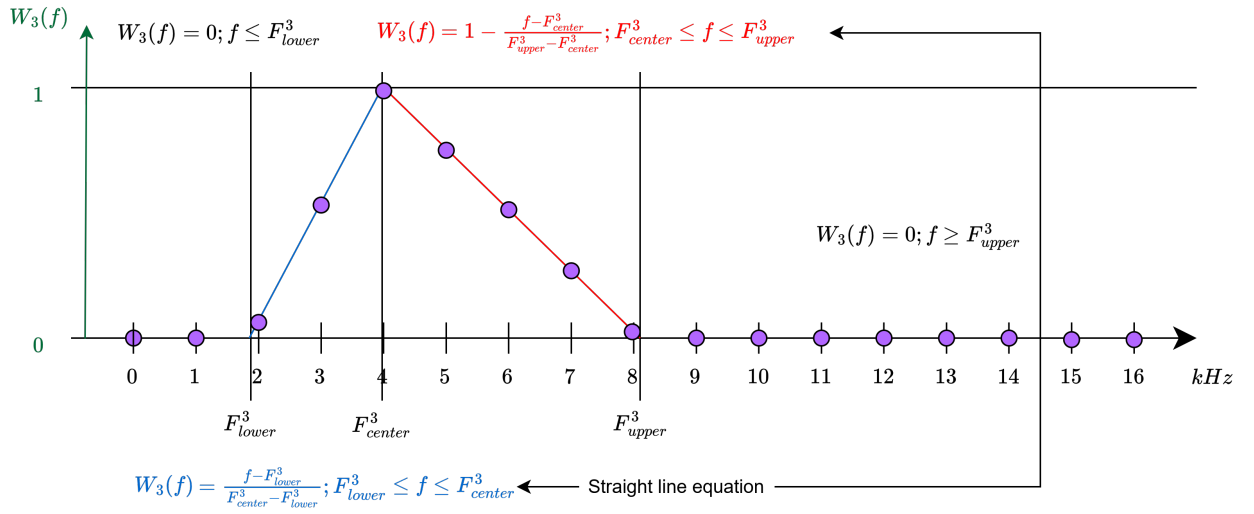


NB - F_{lower}^4 is not at 4kHz, it is at f_3 or 3.996kHz

Note that f_{min} and f_{max} has been indexed as f_0 and f_{M+1}/f_5 respectively. This has been done to complete the triangle filters, W_0 and W_4 following the aforementioned pattern. The weight of the filter ranges from 0 to 1 with 1 occurring at F_{center}^i and 0 at both F_{lower}^i and F_{upper}^i .

Do note how filters centered at higher frequencies cover more linear frequency bins when compared to filters at lower frequencies. This is the logarithmic relationship between linear frequency and Mel frequency as we have seen before.

Lets look at one of the filters in depth –



Follow the color codes for the straight line equations.

The equations can be generalized for any filters as –

$$W_i(f) = \begin{cases} 0, & f \leq F_{lower}^i \\ (f - F_{lower}^i)/(F_{center}^i - F_{lower}^i), & F_{lower}^i \leq f \leq F_{center}^i \\ 1 - (f - F_{center}^i)/(F_{upper}^i - F_{center}^i), & F_{center}^i \leq f \leq F_{upper}^i \\ 0, & f \geq F_{upper}^i \end{cases}$$

Also note that the equation although defined as a continuous function will only be used against the discrete frequency values and so W_i will essentially be an array. For example,

$$W_3 = [0, 0, 0.095, 0.548, 0.999, 0.759, 0.518, 0.277, 0.037, 0, 0, 0, 0, 0, 0, 0]$$

When all the filters are stacked together, we get a matrix W such that,

$$W = \begin{bmatrix} W_1 \\ W_2 \\ W_3 \\ W_4 \end{bmatrix}_{M \times \text{len}(f)}$$

This matrix, W is called the Mel filter-bank.

Bringing it all together

We have created the filter banks that uses nearby information to predict the missing frequency information. How do we apply it now?

Once we have the W matrix, Mel conversion is a simple matrix multiplication. Spectrogram is generally represented as $||X(f, t)||$ which is a 2D matrix of dimension $\text{len}(f) \times \text{len}(t)$. Mel spectrogram, on the other hand, is meant to be a 2D matrix represented as $||X(m, t)||$ of dimension $M \times \text{len}(t)$. This is achieved by simply doing $W @ ||X(f, t)||$ which results in a matrix multiplication between dimension $M \times \text{len}(f)$ and $\text{len}(f) \times \text{len}(t)$ resulting in our desired matrix of dimension, $M \times \text{len}(t)$.

This is all good and fun, but for the exam, how can we do this matrix multiplication? The answer is you can't. You have to find individual cell values through dot product –

$$||X(m = m_0, t = t_0)|| = W_{m_0} \cdot ||X(f, t = t_0)||$$

Do note that both W_{m_0} and $||X(f, t = t_0)||$ have dimensions of $1 \times \text{len}(f)$.

You should get a better idea of the topic by solving the assignment.

Final takeaways

- Mel spectrogram is a type of weighted histogram.
- Beyond all its use cases, Mel conversion can also be used as a dimension reduction technique as multiple frequency info is lumped into a single Mel band.
- Mel spectrogram is always represented in dB. So in case your spectrogram is not in dB, convert it to dB before starting to work on it.
- Although we have studied triangular filters and it is the most common applied Mel filter, there are other variants as well.